Kagan Tam

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science, Nanotechnology Engineering — *Sep 2026 – Present*

Iroquois Ridge High School — Oakville, ON
Team Captain and MVP, 2026 Championship-Winning Volleyball Team — *Sep 2022 – June 2026*

Experience

Formula Nano Design Team — September 2026 – Present
Materials Research and Design Engineer — *Waterloo, ON*

- Developed and refined mechanisms for a custom probing tool, streamlining microfabrication testing procedures to make device characterization more accessible and computationally efficient.
- Executed quantum chemical simulations using ORCA and Density Functional Theory (DFT) to evaluate molecular compatibility, binding affinities, and reaction energetics between candidate chemical compounds.
- Modeled reaction pathways and electronic structures to predict chemical synthesis viability, guiding material selection and accelerating experimental prototyping for nanoscale fabrication.

Research and Design Engineer — April 2024 – September 2025
Shuswap Lake Golf Course — *Blind Bay, BC*

- Spearheaded mechanical design from initial concept to final technical sign off, through a comprehensive 20-page engineering report detailing 3 distinct SolidWorks CAD designs across 5+ iterations between each design.
- Engineered a 3-tier solution matrix balancing cost, durability, and mobility: a heavy-duty permanent installation for maximum lifecycle longevity, a cost-optimized economy build, and a mid-tier modular design engineered for rapid portability.
- Conducted trade-off analyses across 30+ material alternatives and Bill of Materials configurations, delivering actionable capital expenditure proposals ranging from $8,000 to $30,000 for stakeholder decision making.

Assistant Volleyball Coach — November 2025 – March 2026
Iroqouis Ridge High School — *Oakville, ON*

- Collaborated with head coaches to design and execute team practices, developing strategic game plans and tactical drills.
- Analyzed player performance data and game footage to optimize team rotations, defensive positioning, and overall match strategy.
- Coordinated equipment setup, scheduling, and logistics to ensure seamless execution of weekly training sessions and club events.

Projects

Visual Impairment Assistive Goggles | *Arduino, C++, Ultrasonic Sensors, Audio Feedback* — August 2025

- Engineered a wearable assistive navigation prototype integrating an Arduino microcontroller and ultrasonic rangefinders to detect forward obstacles in real time.
- Programmed embedded C++ firmware to convert time-of-flight echo pulses into accurate distance measurements, filtering out sensor noise and false triggers during movement.
- Implemented a dynamic audio feedback loop via an integrated earpiece, translating obstacle proximity into real-time auditory cues to guide safe spatial navigation.

Technical Skills

Languages: Python, C++ (Arduino), HTML/CSS
Software & CAD: SolidWorks, Git, Web Integration Tools
Hardware & Prototyping: Microcontrollers, Ultrasonic Sensors
Engineering Practices: Technical Reporting, Material Evaluation, Design Iteration